

RESEARCH PRESENTATION

Deep Learning Statistical Arbitrage

Reproducing Results and Investigating Trading Costs

Based on Guijarro-Ordóñez, Pelger & Zanotti (2022)

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Outline

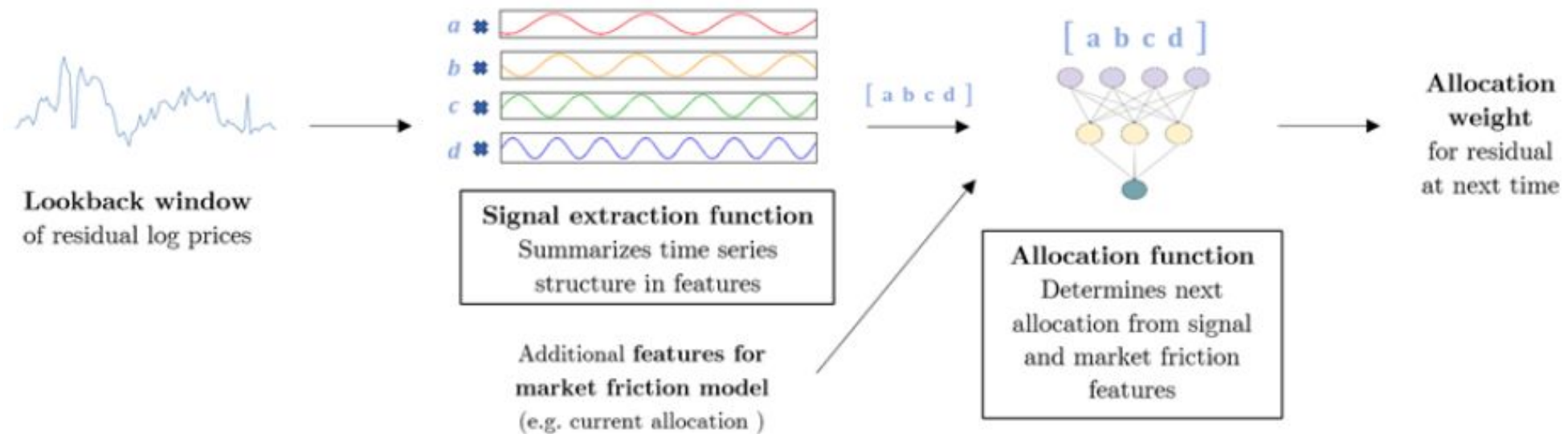
- 01 Problem
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Problem

Statistical Arbitrage: Core Idea

- Trade temporary deviations between similar assets.
- Replace pairs trading with residual portfolios.
- **Question:** can a neural model extract better trading signals?

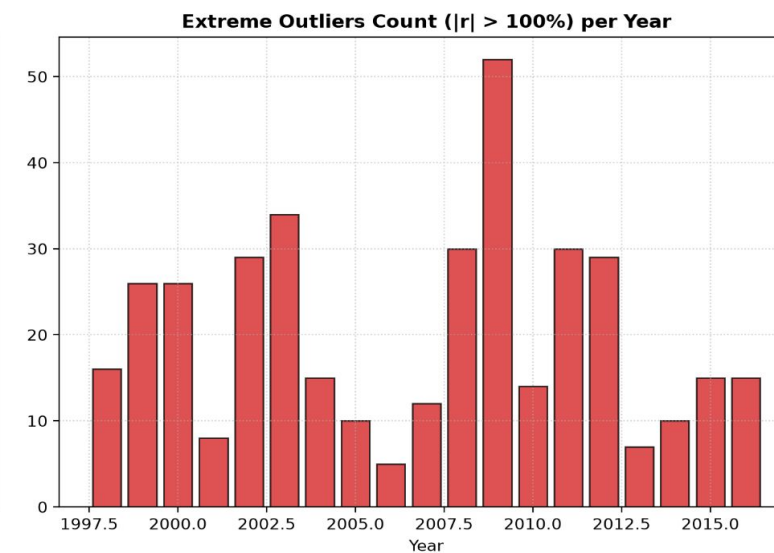
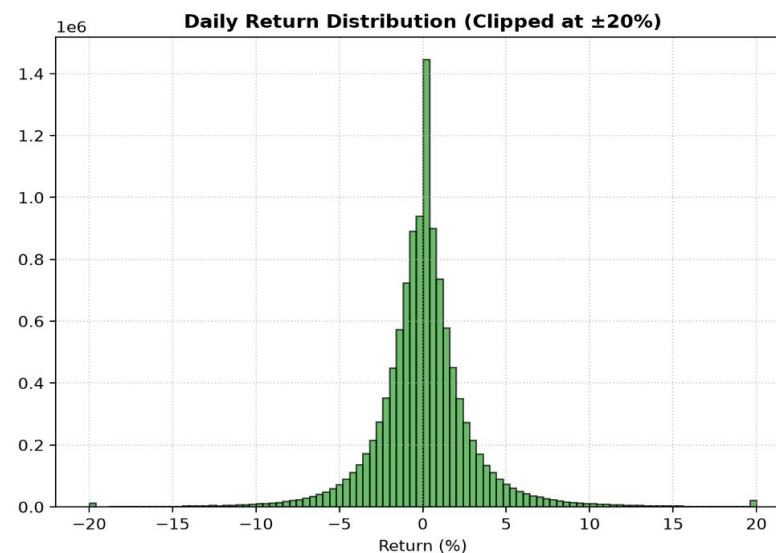
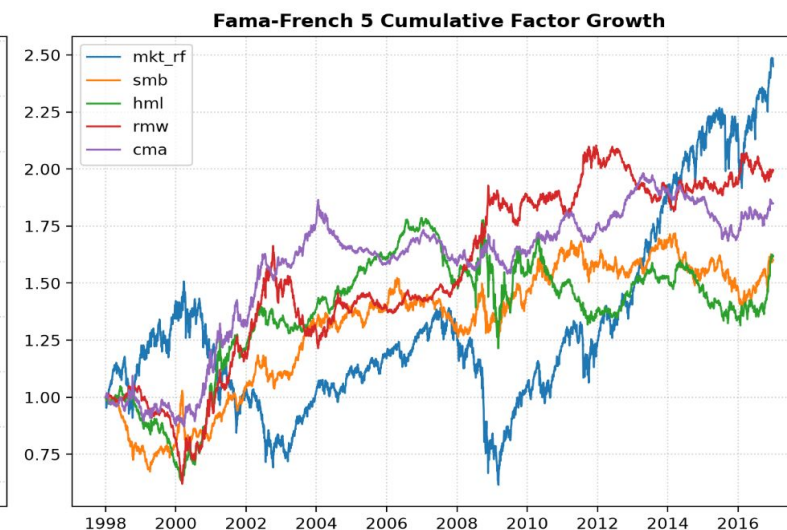
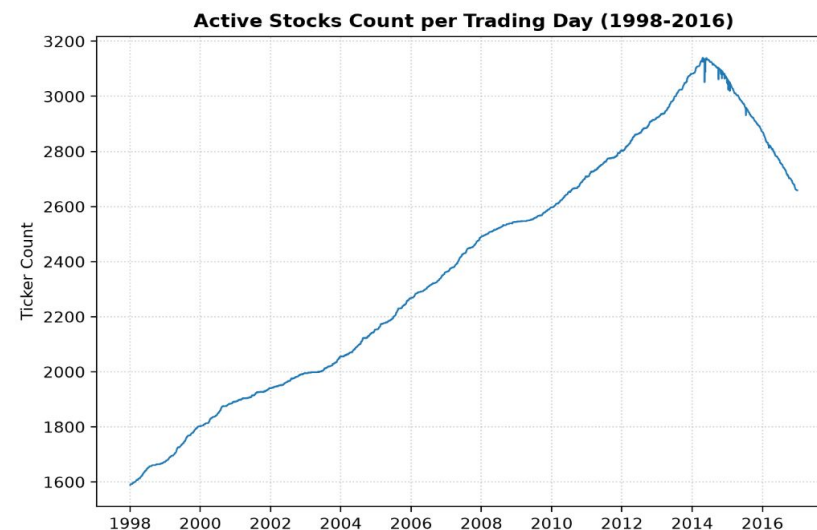


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Method

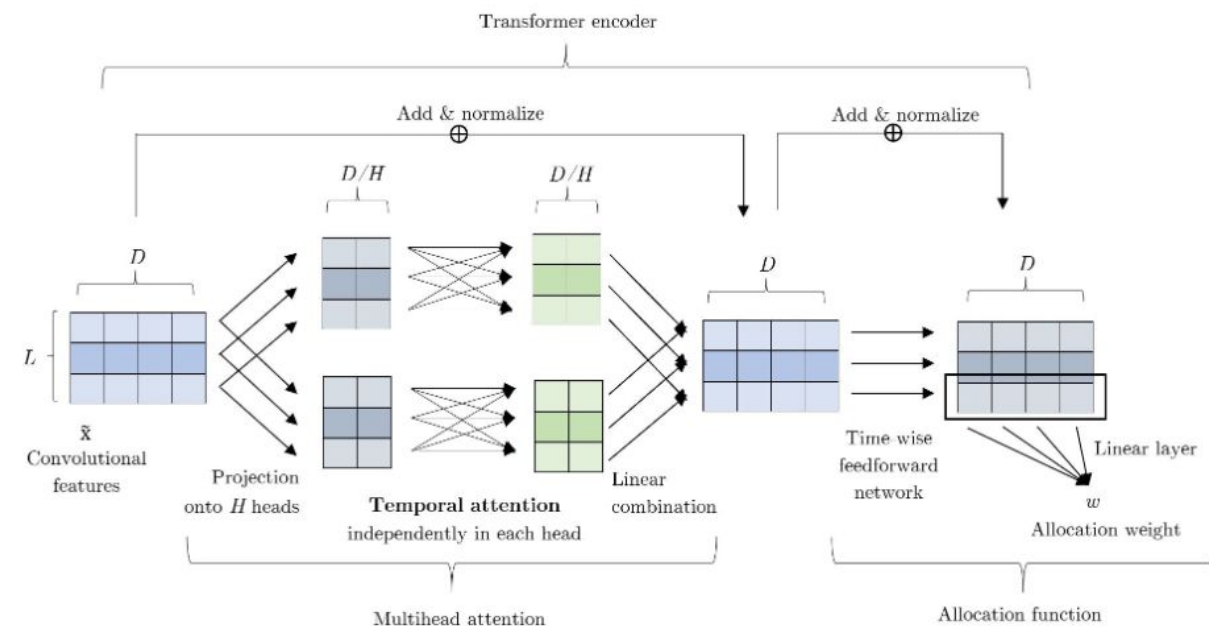
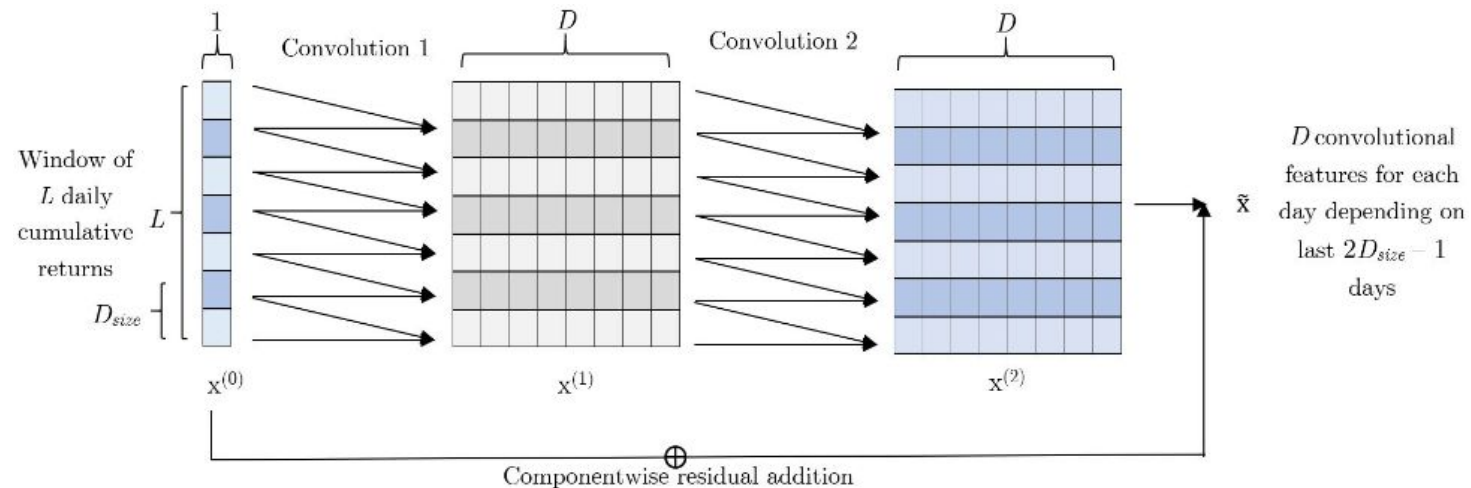
From Returns to Residuals

- Stock returns $\rightarrow$ factor component + residual
- FF5 and PCA5 residuals
- Rolling estimation, no look-ahead



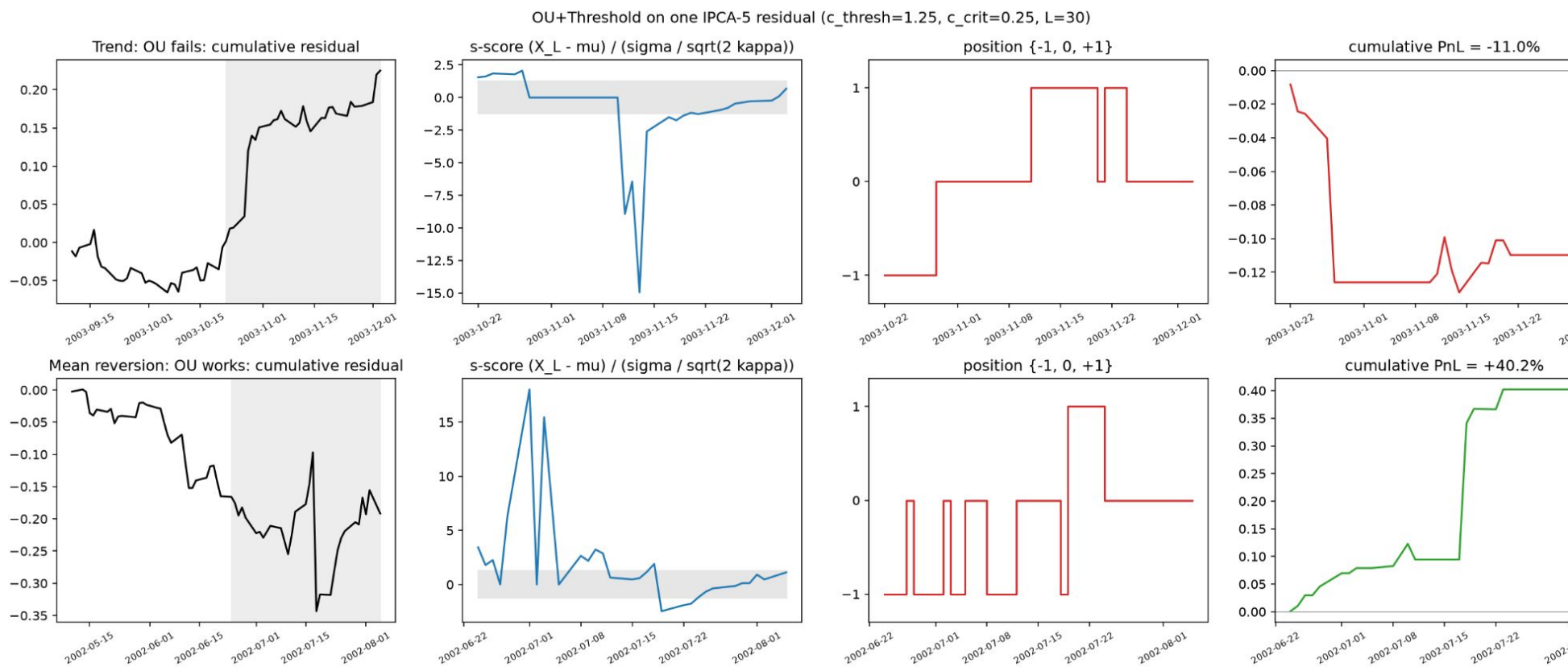
CNN + Transformer Signal Model

- 30-day cumulative residual window
- CNN extracts local patterns
- Transformer combines them over time
- Allocation head scores each stock; scores are normalized into portfolio weights.



OU + Threshold Baseline

- Classical mean-reversion model
- Fit OU process to residual path
- Long cheap residuals, short rich residuals
- Benchmark against CNN+Transformer



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Results & Reproduction

Main Replication Result

- Model ranking is reproduced: CNN+Transformer > Fourier+FFN > OU+Threshold.
- Our Sharpe ratios are lower than the paper's.
- PCA5 performs better than FF5.

Gross annualized Sharpe ratio (higher is better)	Our FF5	Paper FF5	Our PCA5	Paper PCA5
CNN+Transformer	1.95	3.21	2.66	3.36
Fourier+FFN	0.94	1.66	1.32	1.98
OU+Threshold	0.15	0.38	0.62	0.73

* Our CNN+Transformer: mean of two single-seed Sharpe ratios; other models: 1 seed. Gross out-of-sample results, before trading costs. Paper values use a different stock universe.

Benchmarks and Official Residuals

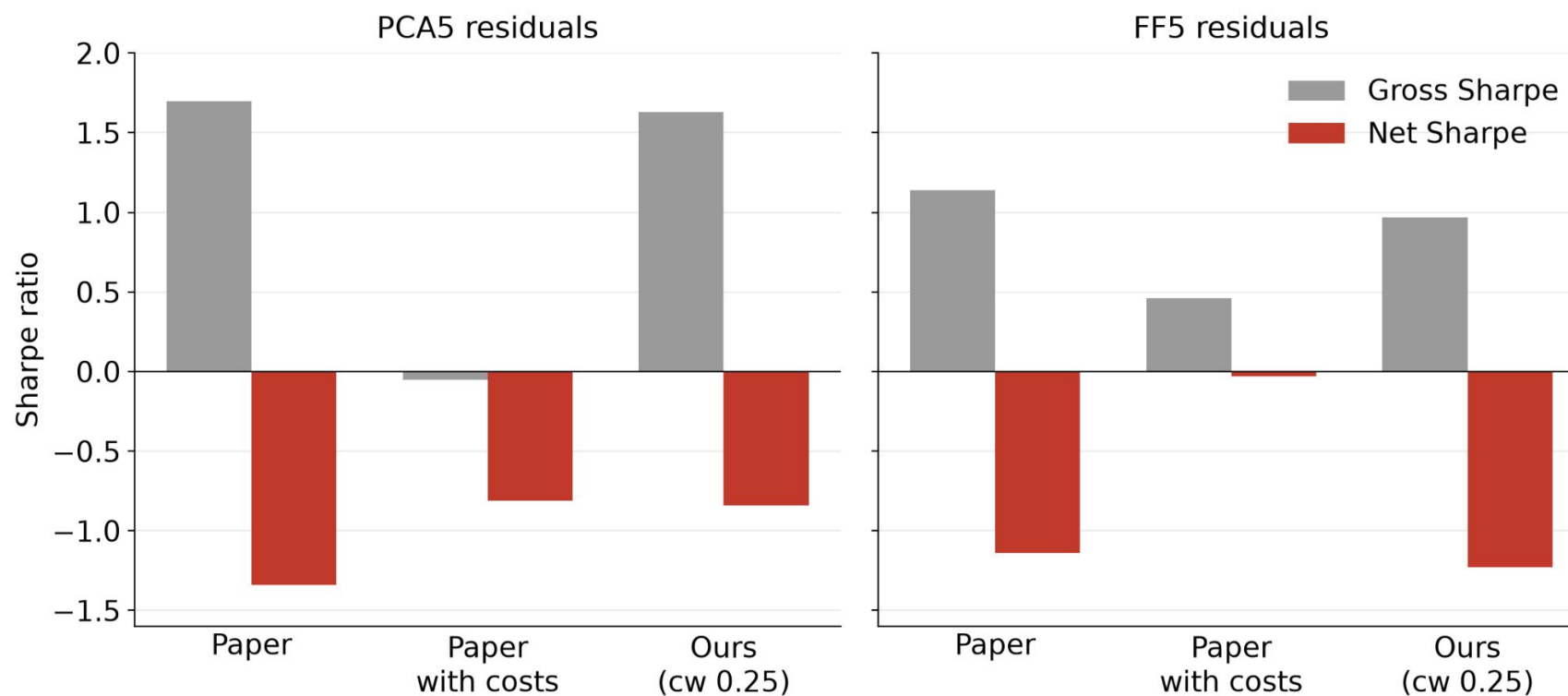
- CNN+Transformer > Fourier+FFN > OU+Threshold
- OU baseline roughly matches paper scale
- Authors' residuals confirm strong gross signal

OOS 2002–2016, authors' residuals: ours / paper

K	Fama-French	PCA	IPCA
0	-0.23/-0.18	-0.23/-0.18	-0.23/-0.18
1	0.01/0.16	0.12/0.21	0.29/0.60
3	0.47/0.54	0.91/0.77	0.71/0.88
5	0.34/0.38	0.93/0.73	0.64/0.97
8	0.04/1.16	0.92/0.87	0.74/0.91

Trading Costs

- Gross Sharpe is positive for the paper and our model.
- After costs, every model is negative on the WIKI test.
- Cost-aware training trades less, but loses the signal.



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Critique

Why Exact Reproduction Is Hard

Paper	Our project
CRSP universe	Quandl WIKI
IPCA + firm characteristics	FF5 / PCA5
Stock-space weights	Residual-space weights
Proprietary data	Public data

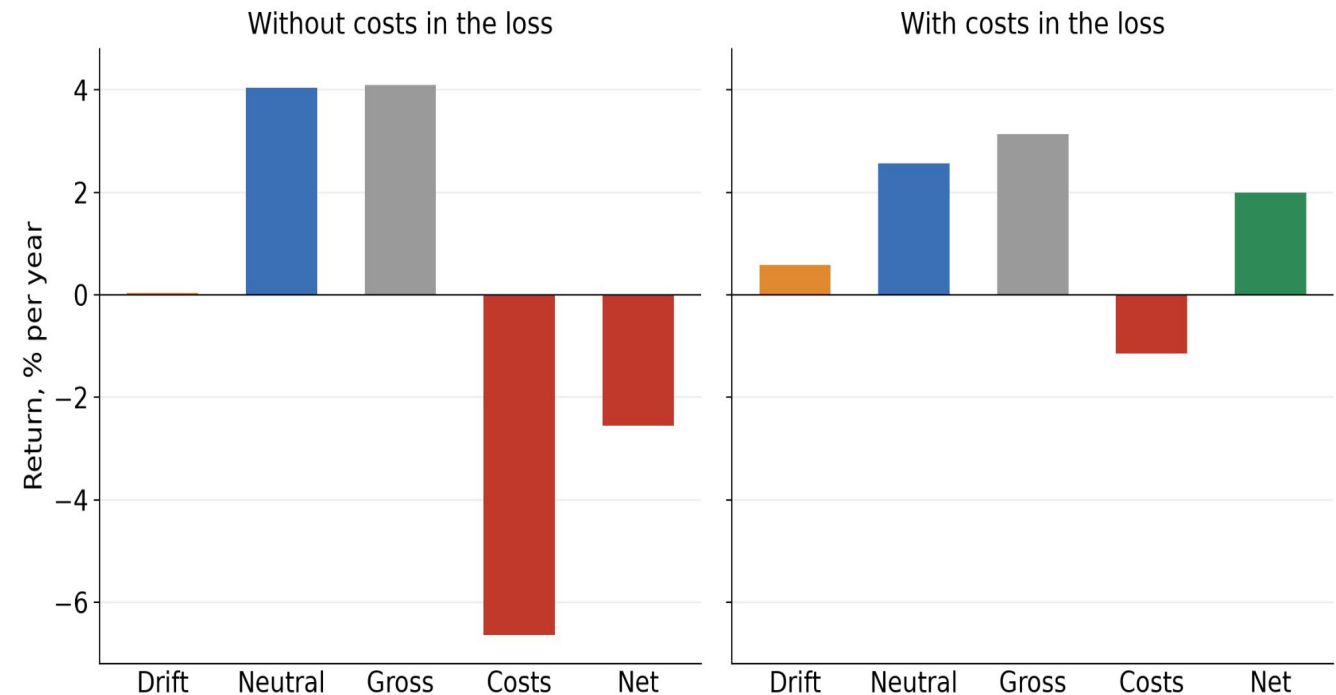
Limitations

- Residuals are traded directly; no Phi mapping to stock weights, so turnover is measured in residual space.
- Our universe (Quandl WIKI, top 500 by dollar volume) is not CRSP; we closed part of the gap with the authors' residuals.
- Simple cost model: 5 bp per unit of turnover + 1 bp/day on shorts; no market impact or capacity limits.
- One test period (2006–2016), 2 seeds per model, many configurations tried (we report deflated Sharpe).

Full list available on Github

Economic Interpretation

- Gross signal is real.
- Net result depends on turnover and costs.
- Cost-aware training may become a net-long book.
- Positive net result is not always pure statistical arbitrage.



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Next Steps

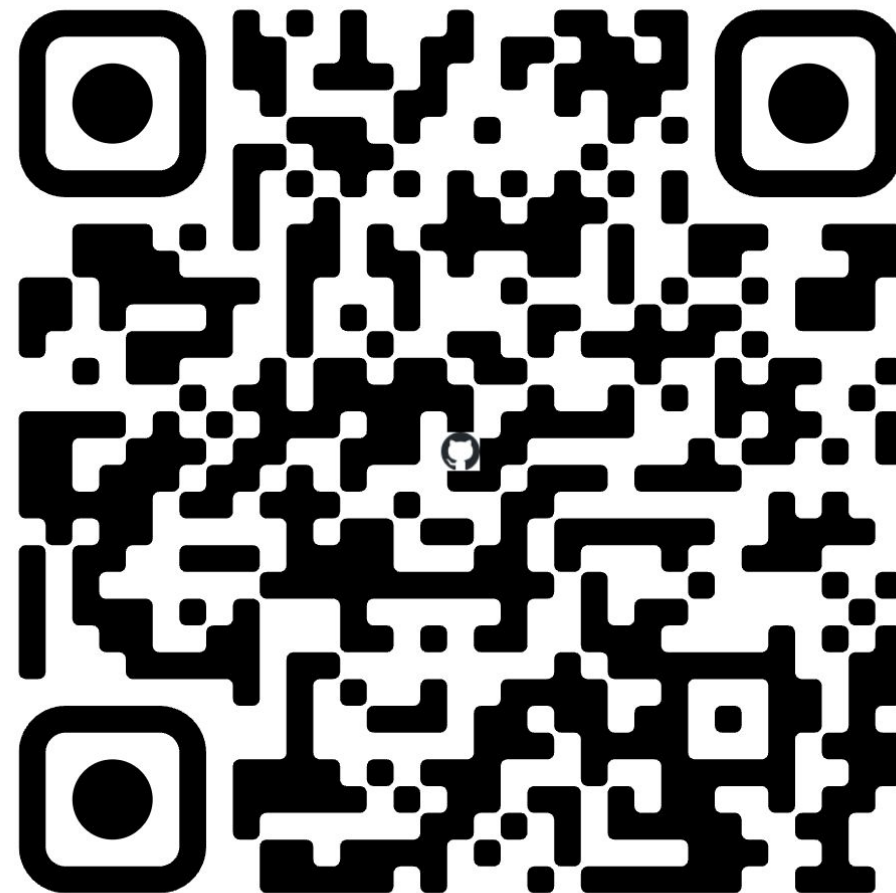
Next Steps and Conclusion

- Reconstruct CRSP-like universe.
- Add firm characteristics and IPCA.
- Evaluate costs in stock space.
- Test on newer OOS periods.

We reproduced the gross statistical signal, but robust net profitability remains unproven.

Thank you

Any Questions?



[Github repo](#)